

Forecasting Household Appliance Energy Consumption

A Comparative Time-Series Study of Benchmark, Statistical, Machine-Learning and Foundation Models

7PAM2032 Time Series Forecasting — Assignment 2 · UCI Appliances Energy dataset · 14-day test horizon

1. Introduction

An ability to accurately predict future levels of domestic energy consumption in the short-term is crucial for successful demand-side management, load scheduling, and incorporation of renewable energy sources. The advances in metering and the developments in the area of machine learning techniques give rise to an important operational question: given several alternative modelling frameworks, some very different from each other, which one should be employed in practice? The answer to this question cannot be provided by a single accuracy measure, as even a slightly better forecast may be biased in a bad way, fail to quantify uncertainty in its predictions, or use unavailable information.

In this report, I conduct a comprehensive analysis of the problem based on the Appliances Energy Prediction dataset published on the UCI repository, which contains readings about appliance energy consumption along with indoor and outdoor climate measurements and an indicator of lighting energy usage collected every ten minutes from January to May 2016. Five classes of models are studied: naive baselines that provide the lowest acceptable level of performance; a seasonal statistical model in target-only (SARIMA) and exogenous (SARIMAX) formulations; a recursive XGBoost model based on engineering features; and Chronos-Bolt and Chronos-2 foundation models without specific training on the dataset.

The questions posed by the brief are answered as they arise in the modelling narrative rather than collected into a list. The report examines the temporal structure of the series and what it implies for modelling; whether the series is stationary and what differencing it requires; which benchmark is strongest and what its dominance reveals about consumer behaviour; whether a seasonal statistical model can beat that benchmark and whether weather information assists it; whether an engineered feature model improves matters and which inputs actually matter; whether a foundation model's accuracy advantage survives scrutiny; which variables are genuinely known when a forecast is made; how the models compare overall and where they fail; and which model a practitioner ought to deploy and why. The recurring finding is that appliance demand is a strongly structured behavioural signal, and that the most accurate model on paper is not the most trustworthy one in practice.

2. Data and Preprocessing

The ten-minute dataset (19,735 records) was sampled into hourly mean records, giving a total of 3,290 records and retaining the daily trend but with reduced noise. The consumption is highly skewed to the right (mean = 97.78 Wh, median = 63.33 Wh, max = 608.33 Wh, skewness = 2.39), with a low base level and very high peaks during the evenings. This is the most difficult part to predict. The final 14 days (336 hours, 13–27 May 2016) form a common test window, forecast by every model from the same training origin to keep the comparison fair and leakage-free.

3. Exploratory Analysis

Figure 1 displays the full hourly time series along with its seven-day moving average, a typical week, and the breakdown of electricity use per hour of the day. The moving average stays almost flat throughout the four months under study, suggesting that there is no long-run trend present; the important feature therefore becomes a periodic one and not a trend. The typical week shows a stable daily pattern, consisting of a low floor of about 45 Wh at night, followed by a slow increase during the day and a marked peak in the evening hours. Appliance demand is therefore a behavioural signal, tied to the hours when a household is active, rather than a smoothly evolving physical quantity. A classical additive decomposition (not shown) confirms the same picture, isolating a large and highly regular 24-hour seasonal component against a gently drifting trend, which directly motivates the seasonal-differencing choice made in Section 5.

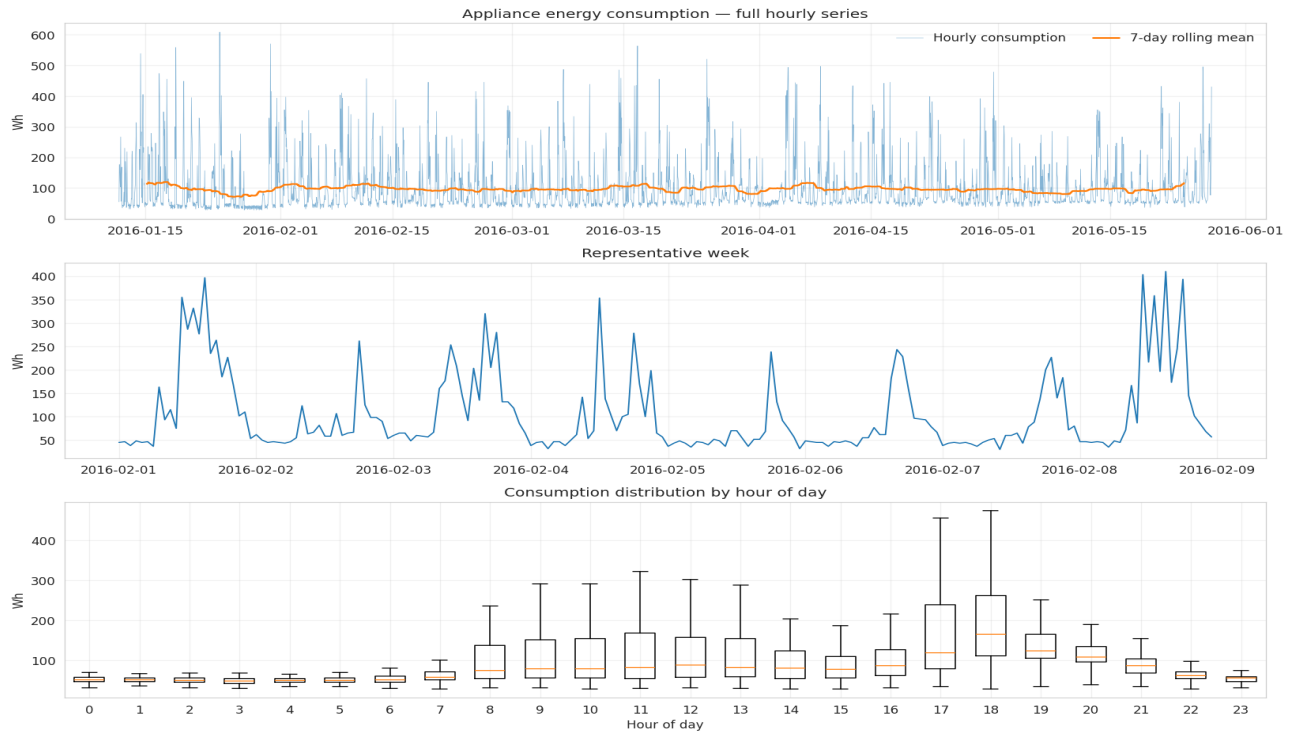


Figure 1. Full hourly series with seven-day rolling mean, a representative week, and the distribution of consumption by hour of day.

Although the dataset supplies indoor-sensor and outdoor-weather covariates, their relationship with demand is weak. Pearson correlations between appliance consumption and the environmental variables are all small: the strongest is with the lighting channel at about 0.26, while outdoor temperature, the humidities and wind speed all fall below roughly 0.2 in absolute value. This early indication that exogenous weather is unlikely to add forecasting value is confirmed repeatedly once the models are fitted, both for SARIMAX in Section 5 and for the conditional XGBoost model in Section 6.

4. Stationarity and Model Identification

Two complementary tests characterise the series. The Augmented Dickey-Fuller test for a unit root shows rejection of the non-stationary property strongly, resulting in a p-value of roughly 8.8×10^{-5} , while on the other hand, the KPSS test with the stationary null hypothesis cannot reject due to a p-value of roughly 0.10. Together they agree that the level series can be treated as stationary, so no ordinary differencing is required. The autocorrelation function nonetheless shows strong, regularly spaced spikes at lags of 24, 48 and 72 hours, confirming that the dependence which remains is seasonal, while the partial autocorrelation function falls away quickly after the first few lags, pointing to a low autoregressive order. These diagnostics identify a model with no regular differencing but a single seasonal difference at 24 hours, together with low-order autoregressive and moving-average terms — precisely the specification adopted in Section 5.

5. Forecasting Design and Evaluation

Model quality is assessed with four complementary measures. The mean absolute error (MAE) reports the average size of the error in watt-hours; the root mean squared error (RMSE) penalises large misses more heavily, so a gap between MAE and RMSE signals a few costly errors; the mean absolute scaled error (MASE) divides the error by that of an in-sample daily seasonal-naïve forecast, so a value below one indicates improvement over that rule and values are comparable across series; and the bias, the mean signed error, exposes systematic over- or under-prediction that magnitude-based metrics conceal. Reading these four together, rather than ranking on MASE alone, is what allows the critical comparison in Sections 8 and 9 to distinguish an accurate model from a trustworthy one.

One further distinction shapes the whole comparison and is fixed before any model is evaluated: what is genuinely known when a forecast is made. Calendar features hour of day, day of week, and month are deterministic and known arbitrarily far ahead, and the target's own past values are known by definition; forecasts built only on these are unconditional and could run in production unchanged. Realised test-period weather and indoor-sensor readings are different: they are only available here because the test period has already occurred, and a live system would have to substitute imperfect forecasts of them, importing further error. Any model consuming such realised

future covariates is therefore a conditional forecast whose apparent accuracy is flattered by information it would not truly possess, and each model below is labelled accordingly so that a fair accuracy comparison is never mistaken for an achievable one.

6. Benchmark Models

Benchmarks set the bar that any more sophisticated method must clear, and the ranking of the five naive forecasts is itself informative. The weekly seasonal naive model is identified as the strongest predictor with MASE 0.798 and MAE 42.63 Wh. Conversely, the daily seasonal naive model produces much weaker results, with MASE 1.628. The average forecast falls somewhere in between, at 0.942. It seems that naive and drift methods work quite poorly with such a volatile series, producing MASE about 4.7 and 5.0, correspondingly. One of the crucial findings is the huge difference between the two seasonal models: predicting one hour using the previous hour of the same day one week ago gives roughly twice as accurate predictions as predicting one hour using the hour of the previous day. Thus, the behavior of appliance demand is structured every week, where weekday dynamics are different from weekend dynamics. This has an immediate design consequence: any model that cannot represent weekly structure begins from a weaker position than a one-line heuristic, and it reframes the covariate question, since the most valuable additional information is the calendar, which encodes the weekly rhythm directly, rather than the weather.

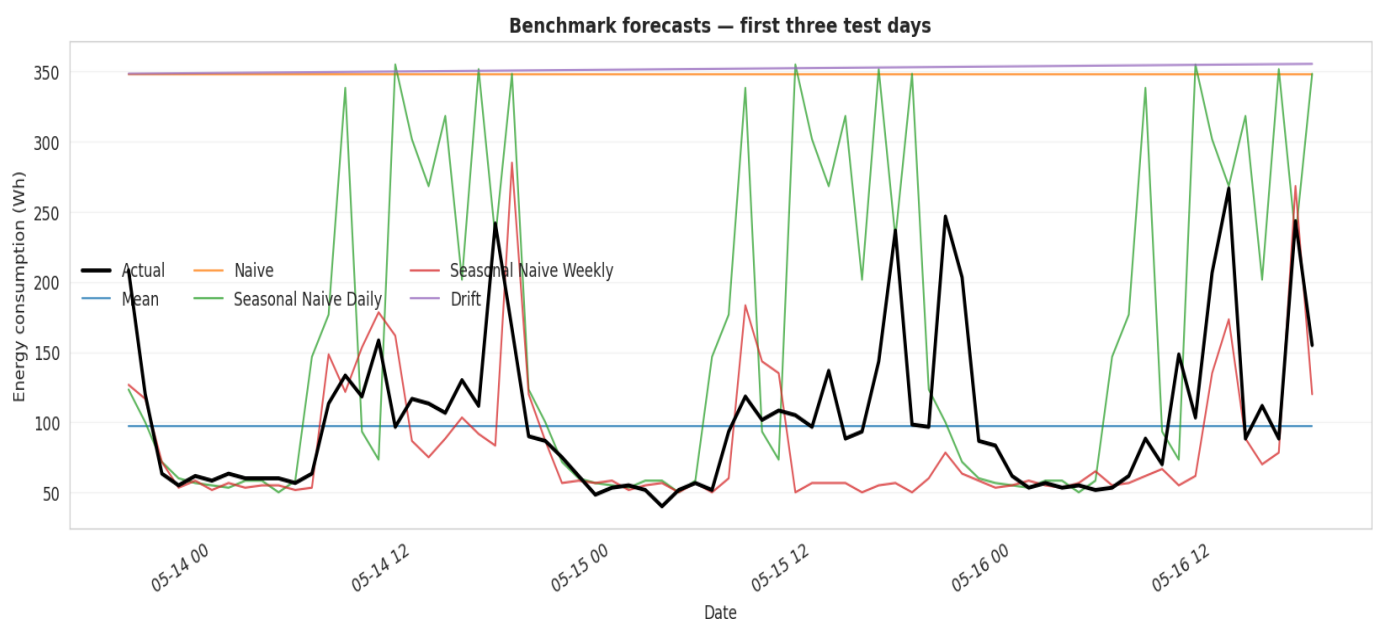


Figure 2. Benchmark forecasts over the first three test days. The weekly rule (red) follows the timing and height of the evening peaks; the daily rule (green) swings sharply between over- and under-prediction.

7. Statistical Modelling: SARIMA and SARIMAX

The statistical stage fits a seasonal ARIMA model at 24 hours. Nonseasonal order grid analysis and archiving is performed by AIC, which generally prefers high-order models; nevertheless, SARIMA (2,0,2)(1,1,1)₂₄ is deliberately chosen as the final model because of its tradeoff between the negligible in-sample performance and increased stability, interpretation, and decrease of overfitting risk. Not using regular differencing and preferring only one seasonal difference is based on the results of Section 4, where seasonal differencing removes daily mean seasonality so that AR and MA terms cover the remaining shorter-range dependencies. The target-only SARIMA reaches a MASE of 0.670, with an MAE of 35.78 Wh and an RMSE of 63.83 Wh, improving on the strongest benchmark by close to sixteen per cent. It clears the weekly seasonal-naive comfortably while producing a calibrated 95 per cent confidence interval that the benchmarks cannot offer. The fit for the 14 days is illustrated in Figure 3 below: The model successfully simulates the daily pattern and the overnight trough. The problem is that the width of the prediction interval becomes wider every following day, and the most significant peaks still occur above the upper limit, demonstrating that the evenings are the hardest to predict.

Figure 3. Target-only SARIMA forecast with 95% confidence interval for 14 days.

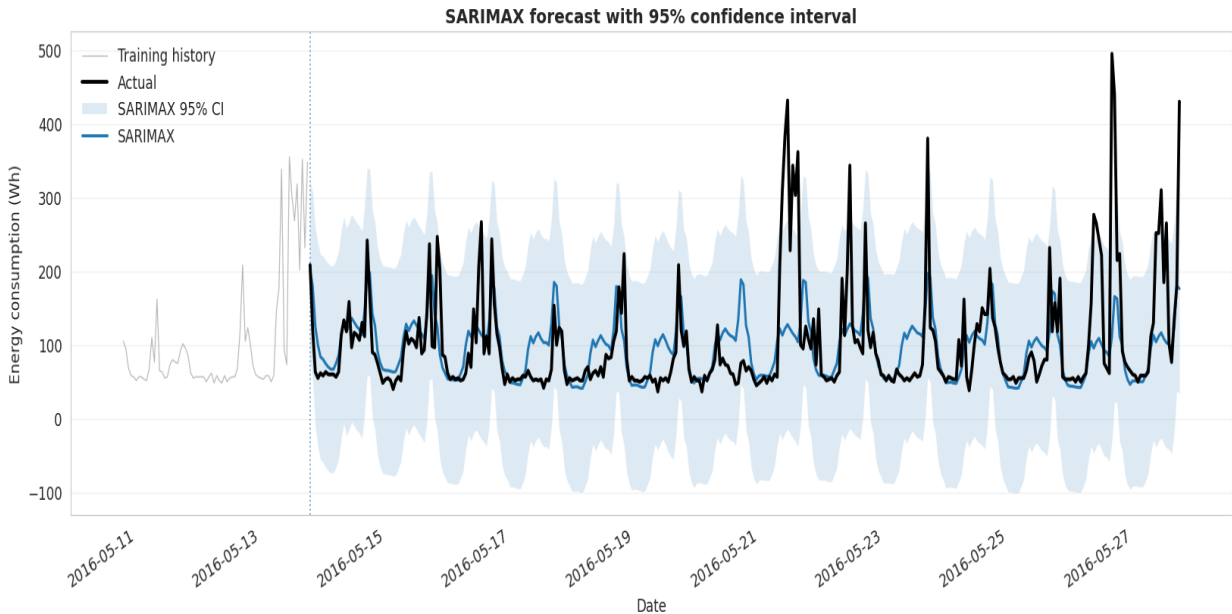


Figure 3. Target-only SARIMA forecast with a 95% confidence interval across the 14-day test window.

The exogenous extension is more revealing than the fit itself. Adding realised weather and calendar terms produces a SARIMAX with a MASE of 0.694, worse than the target-only SARIMA, not better, despite relying on conditional information the honest model refuses to use. The weak correlations noted in Section 3 explain the outcome: injecting low-signal predictors mainly adds parameters and noise, and the model generalises slightly less well. The gain over the benchmark thus comes from capturing seasonal structure, not from the weather, and parsimony beats the larger conditional specification here.

Residual diagnostics confirm the target-only model is adequately specified. In Figure 4 the residual autocorrelation function shows no material structure beyond the shortest lags, the residual distribution is centred and roughly symmetric, and the normal quantile plot is close to the reference line through the body of the distribution, with heavier tails driven only by the rare extreme hours. Ljung–Box tests at lags 24 and 48 do not reject the white-noise hypothesis ($p > 0.05$), so the temporal structure the model was intended to capture has been absorbed.

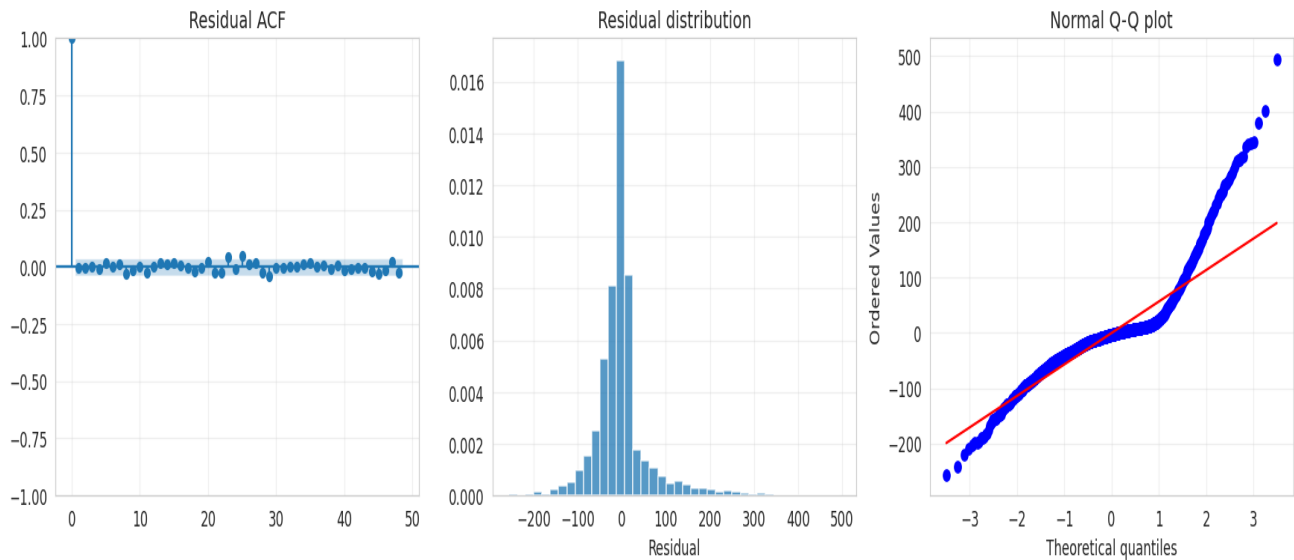


Figure 4. SARIMA residual diagnostics: residual ACF, distribution and normal Q–Q plot.

8. Feature-Based Modelling: XGBoost

The forecasting exercise is then framed as a regression task and handled through the use of a recursive XGBoost model. Each observation is characterized through a predictor row containing autoregressive lags at 1, 2, 3, 6, 12,

24, 48, and 168 hours; rolling mean and standard deviation calculated across 3, 6, 12, 24, and 168-hour windows; and cyclical encoding of hour of day and day of week features. The forecasting exercise takes place recursively, meaning that the predicted values are used as input lags in the subsequent step. Thus, the test-period actuals cannot be used by the model, and the multi-step forecasting problem integrity is retained.

The model performs slightly better than the benchmark, scoring a MASE of 0.763 and an MAE of 40.78 Wh slightly better than the weekly forecast and still lower than SARIMA performance. As can be seen from Figure 5, permutation importances show that the most recent lag feature is significantly more important than all other features, about twice as much as the second one the hour-of-day cyclical features. Also, there are some short lags and not much else. In other words, the model rediscovers two results obtained during the previous exercise that the information is contained in the recent past and in the time of day.

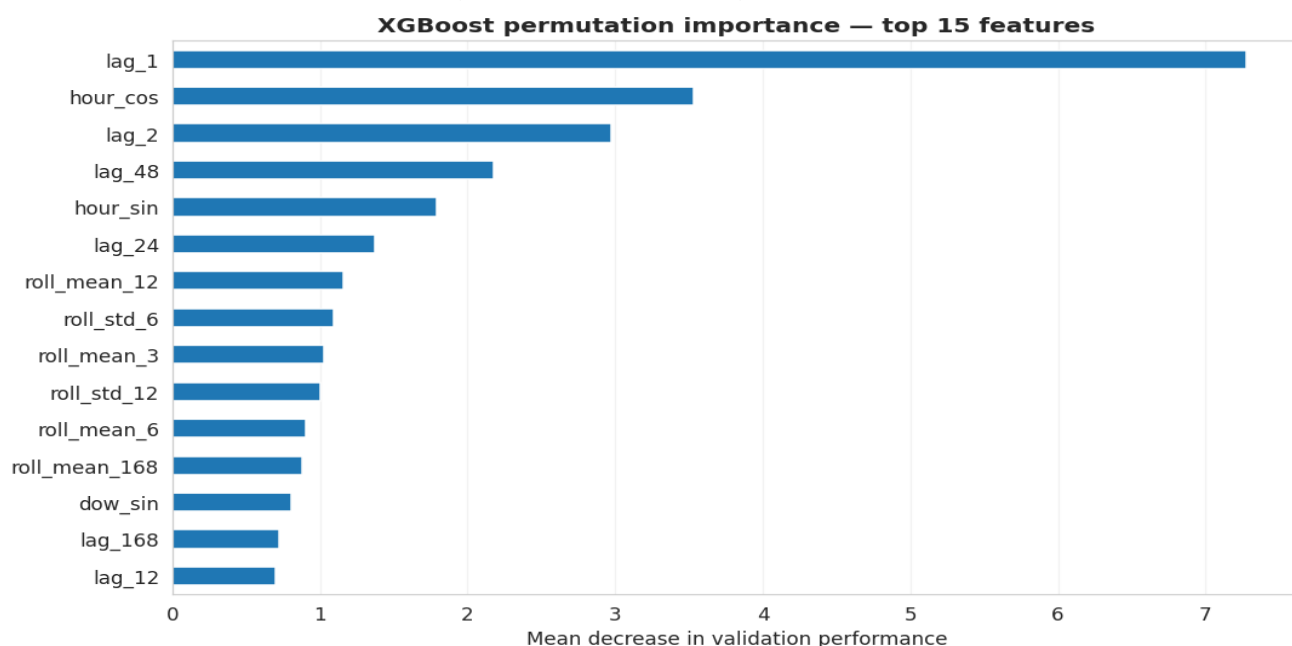


Figure 5. XGBoost permutation importance for the fifteen strongest features; the most recent lag is roughly twice as influential as any other input.

The second XGBoost model, which takes into account indoor-sensor and weather channels, provides the most serious warnings in this study. The MASE value rises from 0.763 to 0.978, thus making the predictive performance almost as bad as in the case of the naïve rule. The grouped permutation importance for the latter helps to understand the process. While the lag group still has the highest importance, the indoor-sensor group is ranked second. Thus, the model is based mostly on the data, which looks informative at the training stage but does not generalize and worsens the forecast, while the weather group has the lowest importance throughout all the time. Thus, the answer to the feature question is clear: adding more features decreases the performance of the model, and only its own past and the calendar can serve as predictors.

9. Foundation Models: Chronos

Finally, the Chronos foundation models are evaluated zero-shot, meaning their pretrained weights are applied with no fitting on this dataset. Univariate Chronos-2 is the strongest model on the primary metric, at a MASE of 0.619 and an MAE of 33.06 Wh, edging past SARIMA by roughly eight per cent. On a metric-only reading, the foundation model wins, and it is tempting to stop there. The fuller picture is less flattering. Chronos-2 carries a large negative bias of -18.98 Wh: it systematically under-predicts, and because the errors it misses are concentrated in the evening peaks, this is the least forgiving direction in which to be wrong for demand planning. SARIMA's bias, by contrast, is only -4.15 Wh, and SARIMA actually records the lowest RMSE of the leading models at 63.83 Wh against 65.51 Wh, so its largest errors are smaller even though its average error is marginally higher. The foundation model also provides no native uncertainty interval and is opaque by construction, and, revealingly, it becomes worse when covariates are supplied: the covariate-informed variant slips to a MASE of 0.650 with an even larger bias of -22.04 Wh. The accuracy advantage is real but narrow, and it is bought at the cost of bias, uncertainty and interpretability rather than being justified by them.

10. Comparative Results and Error Analysis

Table 1 collects the headline metrics for every model and marks each as unconditionally deployable from the forecast origin or conditional, reliant on realised future covariates. Chronos-2 leads on MASE and MAE, SARIMA leads on RMSE and on bias among the accurate models, and the conditional variants of both SARIMAX and XGBoost fall below their honest counterparts, the clearest possible sign that the additional information is not paying its way.

Model	Type	MAE	RMSE	MASE	Bias
Chronos-2 (univariate)	Unconditional	33.06	65.51	0.619	-18.98
SARIMA (target-only)	Unconditional	35.78	63.83	0.670	-4.15
Chronos-2 + covariates	Conditional	34.72	70.38	0.650	-22.04
SARIMAX (+ exogenous)	Conditional	37.05	64.03	0.694	-3.72
XGBoost (lags + calendar)	Unconditional	40.78	66.12	0.763	+4.05
Seasonal-naïve (weekly)	Benchmark	42.63	79.29	0.798	-10.82
Mean	Benchmark	50.32	74.91	0.942	-3.11
XGBoost + sensors/weather	Conditional	52.25	71.22	0.978	+20.80
Seasonal-naïve (daily)	Benchmark	86.96	129.23	1.628	+64.01

Table 1. Test-set performance over the 14-day horizon (errors in Wh), ordered by MASE. Lower is better for MAE, RMSE and MASE; bias nearest zero is best.

Figure 6 overlays the leading models on the actual series. All capture the broad daily rhythm and the weekday-to-weekend shift, but the tallest spikes, the hours above roughly 300 Wh, are under-shot by every model, most visibly by Chronos-2, whose curve sits consistently below the peaks and so accumulates the negative bias identified earlier. SARIMA and its interval track the body of the series closely, and the weekly seasonal-naïve remains a respectable reference throughout.

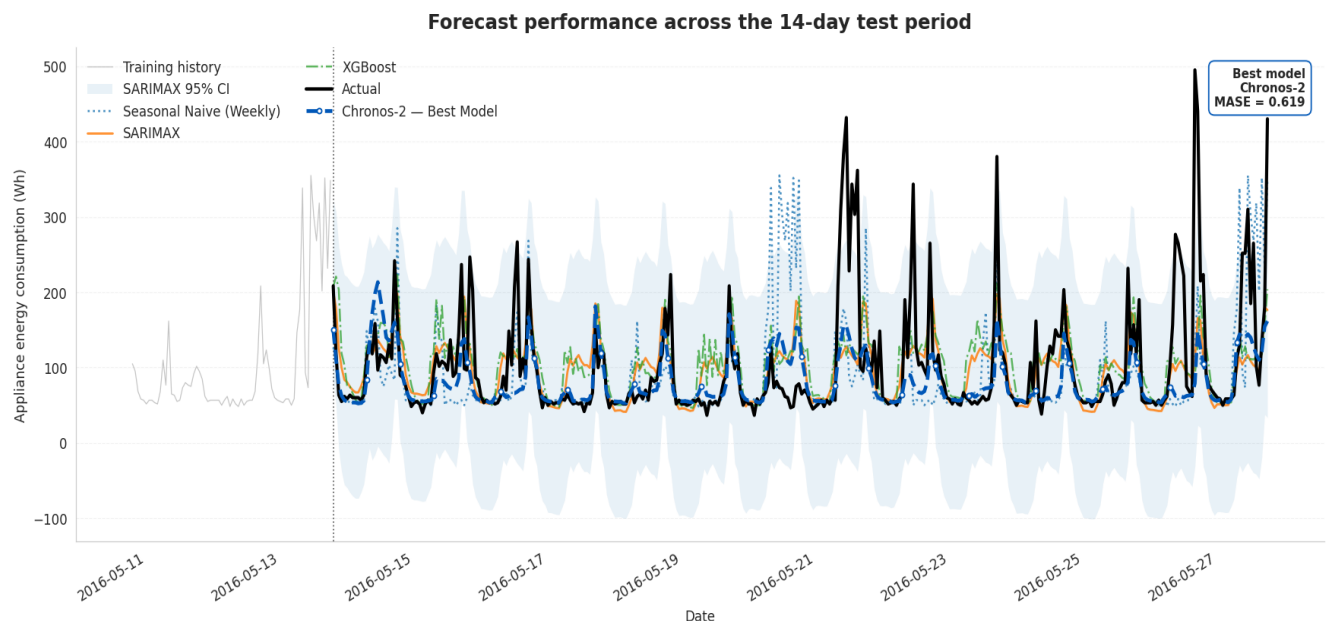


Figure 6. Forecast performance across the test period, with the actual series shown against the leading models and the SARIMA interval.

Figure 7 pairs the MASE ranking with the distribution of absolute errors for the strongest benchmark and the leading models. The ranking confirms the ordering in Table 1, and the boxplots add nuance: Chronos-2 shows the lowest median absolute error and the tightest spread, so its central accuracy is genuinely strong. Its weakness is one of direction rather than magnitude the systematic under-prediction is a bias that the absolute-error boxes cannot display, which is exactly why judging on the error distribution alone would miss the flaw and why the signed bias in Table 1 is essential to a fair verdict.

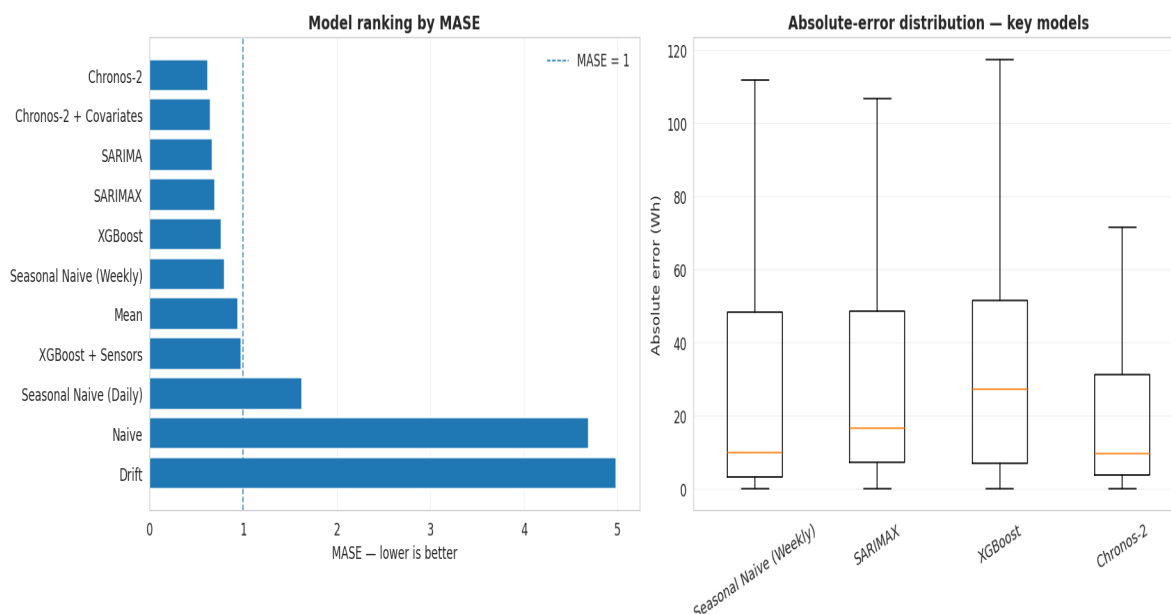


Figure 7. MASE ranking (left) and absolute-error distributions (right) for the strongest benchmark and the leading models.

11. Discussion: Information, Trade-offs and Deployment

The distinction between conditional and unconditional forecasts proves decisive when the results are read as a whole. Only the calendar features and the target's own historical lags are genuinely known at the forecast origin; realised weather and sensor readings are not, and every model relying on them is conditional. The evidence that this matter is direct: the unconditional SARIMA, at a MASE of 0.670, is only about two and a half per cent worse than the conditional SARIMAX at 0.694 and the conditional model is in fact the poorer of the two despite its access to perfect future weather. The extra information does not merely fail to help; it slightly hurts. Honest models built only on the calendar and past consumption are therefore strongly preferable, being genuinely deployable while losing nothing of substance.

This leads to a recommendation that follows from the full metric picture rather than the headline ranking. Chronos-2 is the accuracy-oriented option, marginally best on MASE and MAE with no tuning, but it under-predicts the peaks that matter, offers no measure of its own uncertainty, cannot be readily interrogated, and is degraded rather than helped by covariates. SARIMA is the balanced option: within about five per cent of Chronos-2 on scaled error, the best of the accurate models on RMSE, effectively unbiased, equipped with a calibrated confidence interval, cheap enough to refit continuously on modest hardware, transparent to an auditor, and dependent only on information available at the origin. For a practical smart-home or utility deployment, SARIMA is accordingly the model to run, because near-zero bias and honest uncertainty matter more for scheduling and capacity decisions than a few per cent of average error; Chronos-2 is worth holding in reserve for settings where maximum point accuracy is the sole objective and its bias and opacity can be tolerated or corrected, while the weekly seasonal-naive remains a valuable monitoring baseline precisely because it is so hard to beat by much. Running through every stage is a single broader lesson: recent consumption history and time-of-day carry almost all of the predictive signal, so effort is better directed towards understanding occupancy patterns and maintaining clean historical data than towards expanding the range of environmental inputs.

Limitations and Future Work

Several limitations qualify these conclusions. The data span only around four months of a single dwelling, so seasonal generalisation cannot be assessed, and the weekly pattern found here may not transfer to other households or times of year. A single fixed test origin is used rather than rolling-origin evaluation, which would give a more robust picture of how the model ranking holds up across different windows. The recursive XGBoost accumulates error over the 336-hour horizon, and the foundation models were applied strictly zero-shot. The most valuable next steps would be rolling-origin evaluation across several historical windows, light fine-tuning of Chronos-2 to test whether its peak under-prediction can be removed without sacrificing accuracy, and the replacement of weak weather inputs with genuine occupancy signals such as motion or carbon-dioxide sensors, which the evidence here suggests would help considerably more than the environmental variables already available.

Conclusion

The seasonal-naive weekly rule poses a tough benchmark due to its superiority, and its superiority suggests that demand exhibits a weekly pattern. A parsimonious SARIMA model outperforms the seasonal-naive rule since it accounts for the seasonality, while additional variables related to weather always fail to increase the performance of such a model. A feature-based model follows a similar hierarchy and predicts that the lags and time-of-day include the most informative features, while a foundation model shows the highest headline accuracy, yet it tends to underestimate peak values and does not account for its own uncertainty. Taking into consideration the multi-faceted nature of evaluation criteria and not just one criterion, it is necessary to conclude that the use of Chronos-2 can be recommended in cases where point accuracy is very important, while a SARIMA model should be applied as an operational model due to more practicality and utility of a forecast which discloses its information source, is unbiased toward peaks, and knows its own uncertainty.

References

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